XIAMEN UNIVERSITY MALAYSIA

Online Quiz 2

Course Code:	COM108			
Course Name:	Engineering Maths (I)			
Question Paper Setter:	Liu Mei Feng			
Academic Session:	2021/09	Question Paper:	A ☐	B ☐
Total No. of Pages:	1	Time Allocated:	2 hours	
Additional Materials:	-			
Apparatus Allowed:	-			

INSTRUCTIONS TO CANDIDATES

1. There are four questions in this paper. Please do ALL questions.

2. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (20 marks)

Evaluate each of the following limits.

(a) $\lim_{z\to\infty} \frac{4z^2+z^6}{1-5z^3}$ [4 marks]

(b) $\lim_{t\to-\infty} \frac{t^2-5t-9}{2t^4+3t^3}$ [4 marks]

(c) $\lim_{x\to-\infty} \frac{2x^4-x^2+8x}{-5x^4+7}$ [4 marks]

(d) $\lim_{s\to0^+} e^{\frac{1}{s}}$ [4 marks]

(e) $\lim_{x\to-\infty} \frac{e^{6x}-4e^{-6x}}{2e^{3x}-5e^{-9x}+e^{-3x}}$ [4 marks]

Question 2 (10 marks)

(a) If $f(x)+x^2[f(x)]^5=34$ and $f(1)=2$, find $f'(1)$. [5 marks]

(b) Find dy/dx by implicit differentiation. [5 marks]

$$x^2-8xy+y^2=8$$

Question 3 (10 marks)

Given the function

$$f(x)=\begin{cases} x+2, & \text{if } x<2 \\ ax^2, & \text{if } x=2 \\ b, & if\ x>2 \end{cases}$$

(a) Find the following limit if $f(x)$ is continuous at $x=2$. [4 marks]

$$\lim_{x\to2^-} f(x)$$

(b) Find the value of a and b that will make $f(x)$ continuous on $(-\infty,\infty)$. [6 marks]

Question 4 (10 marks)

Evaluate the derivative of the following function.

(a) [5 marks]

$$f(x)=(e^x+x^2)\left(\frac{1}{\sqrt{x}}+\sqrt[3]{x}\right)$$

(b) [5 marks]

$$f=(\sin x)^{\ln x}$$

- END OF PAPER -